8. Sort a given N integer elements using Heap Sort technique and compute it's time taken.

```
#include<stdio.h>
# include<time.h>

         max
void heapify (int arr[], int n, int i){
        int largest = i;
        int left = 2*i+1;
        int right = 2*i+2;

        if(left <n && arr[left]> arr[largest])
            largest = left;

        if (right <n && arr[right] > arr[largest])
            largest = right;

        if(largest != 1){
            int temp= arr[i];
            arr[i]= arr[largest];
            arr[largest] = temp;

        max heapify(arr, n, largest);
        }
}

      max
void heapSort(int arr[], int n){
    for(int i=n/2-1; i>=0; i--){
        maxheapify(arr, n, i);
    }
```

Scanned with OKEN Scanner

```
    for(int i=n-1; i>0; i--){
        int temp = arr[0];
        arr[0] = arr[i];
        arr[i] = temp;
        maxHeapify(arr, i, 0);
    }
}

void minHeapify(int arr[], int n, int i){
        int smallest = i;
        int left = 2*i+1;
        int right = 2*i+2;

        if (left <n && arr[left] <arr[smallest])
            smallest = left;

        if(right< n && arr[right] <arr[smallest])
            smallest = right;

        if (smallest != i){
            int temp = arr[i];
            arr[i] = arr[smallest];
            arr[smallest] = temp;
            minHeapify(arr, n, smallest);
        }
}

void minHeapSort(int arr[], int n){
    for(int i=n/2 -1; i>=0; i--)
        minHeapify(arr, n, i);
```

```
    for(int i = n-1; i>0; i--){
        int temp = arr[0];
        arr[0] = arr[i];
        arr[i] = temp;
        minHeapify(arr, i, 0);
    }
}

void printArray (int arr[], int n){
    for(int i= 0; i<n; i++)
        printf("%d", arr[i]);
    printf("\n");
}

int main(){
    int n, choice;
    printf("Enter number of elements:");
    scanf("%d", &n);
    int arr[n];

    printf("Enter elements:\n");
    for(int i=0; i<n; i++)
        scanf("%d", &arr[i]);

    printf("1.MaxHeapSort (Ascending)\n");
    printf("2. MinHeapSort (Descending) \n");
    printf("Enter choice: ");
    scanf("%d", &choice);
    clock_t start = clock();

    if (choice == 1)
```

```
        maxHeapSort (arr, n);
    else
        minHeapSort(arr, n);
    clock_t end = clock();

    printf("Sorted array:\n");
    printArray(arr, n);

    double time_taken = ((double)(end - start))/CLOCK_PER_SEC;
    printf("Time taken: %f seconds\n", time_taken);
    return 0;
}
```

O/P:

```
Enter number of elements: 5
Enter element:
42 54 64 23 21
1. Max Heap Sort
2. Min Heap Sort
Enter choice: 1
Sorted Array:
21 23 42 54 64.

Enter number of elements: 5
Enter element:
42 54 64 23 21
1. Max Heap Sort
2. Min Heap Sort
Enter choice: 1
Sorted Array.
64 54 42 23 21
```

27/11